

zakon o ohranitvi energije:

$$W = \Delta E$$

delo sile
izven sistema
na sistem

spr. energije
sistema

če opazujemo sistem in nič ne naredimo

$$\underline{\Delta E = 0}$$

① Delo

Poskusi

1.) $\varphi = 0$

$$W = +Fd$$

2.) $\varphi = \pi$

$$W = -Fd$$

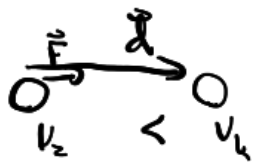
3.) $\varphi = \frac{\pi}{2}$

$$W = 0 \cdot Fd$$

$$\Rightarrow W = Fd \cos \varphi = \vec{F} \cdot \vec{d}$$

$$[J = Nm = \frac{kg \cdot m^2}{s^2}]$$

② Kinetična energija



$$W = \Delta E = E_2 - E_1$$

$$F = ma \quad v_1^2 = v_2^2 + 2ad$$

$$W = Fd = m \frac{v_1^2 - v_2^2}{2} = \frac{1}{2} m v_1^2 - \frac{1}{2} m v_2^2$$

$$W_k = \frac{1}{2} m v^2$$

③ Potencialna energija

$$W = \Delta E = E_2 - E_1$$

$$W = W_{p2} - W_{p1}$$

$$= Fd$$

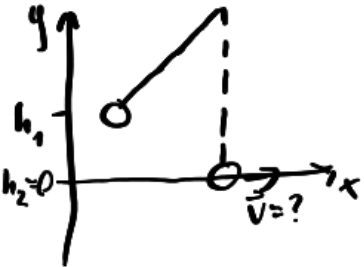
$$mgd = F = mg$$

$$m g (h_2 - h_1)$$

$$W = m g h_2 - m g h_1$$

$$\boxed{W_p = m g h}$$

Nihanje

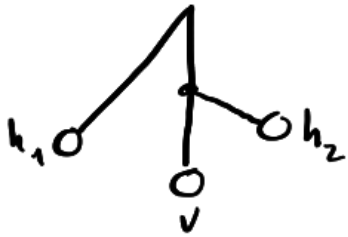


$$\Delta E = W = 0$$

$$E_2 = E_1$$

$$m g h = \frac{1}{2} m v^2$$

$$v = \sqrt{2 g h}$$



$$\Delta E = 0$$

$$h_1 = h_2$$

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